Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.3 – Series Circuits

Unit 2: Electricity & Energy

The question: *in a loop with several bulbs, is the current the same everywhere — and does each bulb get the full push from the cell, or does the push get shared out?* Today's circuits all have one thing in common: **one loop, no junctions.** That single fact decides both rules below.

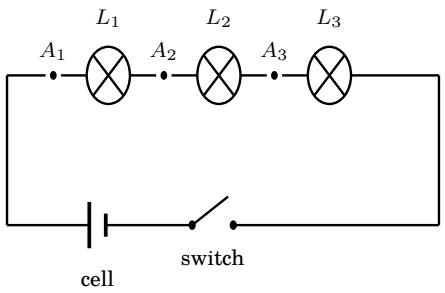
Learning intentions

All students will state and apply the **current rule** and the **voltage rule** for series circuits.
Most students will explain what adding another bulb does to the others, without a number.
Some students will calculate a missing voltage using both rules together.

Part 1 — The current rule

In a series circuit, the current has **only one path** round the loop — there is nowhere else for it to go. So:

The current is the SAME at every point in a series circuit.



1) **Predict** the readings at A_2 and A_3, if A_1 reads **0.4 A**.

Model answer: *Both read 0.4 A — the current is the same everywhere in a series loop.*

Build it

Build the three-bulb circuit and place your ammeter at A_1, then A_2, then A_3 in turn. Record all three.

$A_1 =$ _____ A $\qquad$ $A_2 =$ _____ A $\qquad$ $A_3 =$ _____ A

Check: all three readings should be the same, within the accuracy of your meter.

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Part 2 — The voltage rule

The cell gives every bit of charge a fixed push — a fixed voltage. That push gets **used up** crossing each bulb in turn, so:

> **The voltages across the components ADD UP to the cell's voltage.**

2) **Calculate** the voltage across L_2, if the cell is **6 V** and L_1 reads **2.5 V**.

Model answer: $6 - 2.5 = 3.5\,V$ *across* L_2 *— the two voltages must add back up to the cell's 6 V.*

Build it

Measure the voltage across the cell, then across L_1 and L_2 only (ignore L_3 for this check).

Cell = _____ V L_1 = _____ V L_2 = _____ V

Check: $L_1 + L_2$ should add up close to the cell's voltage — not exactly, because a real circuit loses a little voltage in the wires and the meters themselves.

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Most students — adding another bulb

3) Explain what happens to the brightness of **every** bulb in the three-bulb circuit if you add a fourth bulb into the same loop. You do not need a number — explain using the current rule.

Model answer: *Every bulb gets dimmer. Because there is only one path, the SAME current flows through all four bulbs — and a fourth bulb in the way means less current can get round the loop at all, so every bulb (not just the new one) sees less current than before.*

Some students — using both rules together

4) Calculate the voltage across L_3 in a series circuit with a **9 V** cell, where $L_1 = 3\,\text{V}$ and $L_2 = 4\,\text{V}$.

Model answer: $9 - 3 - 4 = 2\,V$ *across* L_3.

Part 3 — On your own

A series circuit has a cell and three bulbs. The ammeter reads **0.2 A**. The cell is **12 V**. L_1 reads **5 V** and L_2 reads **3 V**.

5) Find the current through L_3, and the voltage across L_3. Show which rule you used for each.

Model answer: *Current through* L_3 *= 0.2 A (current rule — same everywhere in series). Voltage across* $L_3 = 12 - 5 - 3 = 4\,V$ *(voltage rule — the three voltages add up to the cell's 12 V).*

HOMEWORK

Old-style fairy lights are wired in series — one loop, all the bulbs. If **one** bulb in the string burns out and its filament breaks, explain what happens to the **rest** of the string, and why — using what you learned about series circuits today.